📊 Dataset Overview Report

Fuel Type Prediction – US Automobile Market Analysis

---

1) Dataset Description

The dataset contains detailed specifications of passenger vehicles in the US automobile market. It includes both technical and design-related attributes that influence vehicle performance, efficiency, and pricing.

The primary objective of this analysis is to predict the **fuel type** (Gas = 0, Diesel = 1) using vehicle characteristics.

---

2) Dataset Size

- **Total Records:** 205 vehicles
- **Total Features:** 26 columns
- **Target Variable:** fueltype
  - Gas = 0
  - Diesel = 1

---

3) Data Structure

The dataset consists of:

🔹 **A. Numerical Features**

Examples include:

- wheelbase
- carlength
- carwidth
- curbweight
- enginesize
- compressionratio
- horsepower
- peakrpm
- citympg
- highwaympg
- price

These variables describe physical dimensions, engine specifications, fuel efficiency, and pricing.

🔹 **B. Categorical Features**

Examples include:

- aspiration
- doornumber
- carbody
- drivewheel
- enginetype
- cylindernumber
- fuelsystem

These variables describe design and structural characteristics of vehicles.

---

4) Data Quality Assessment

After performing initial data inspection:

- ✅ No missing values were found.
- ✅ No duplicate records were detected.
- ✅ Data types were correctly structured (numerical and categorical clearly separated).

This indicates the dataset is clean and suitable for modeling after encoding and scaling.

---

5) Target Variable Distribution

The dataset shows class imbalance:

- Gas vehicles form the majority class.
- Diesel vehicles represent a smaller proportion of the dataset.

To address this imbalance, **SMOTE (Synthetic Minority Over-sampling Technique)** was applied on the training dataset before model building.

---

6) Business Relevance of Dataset

The dataset provides critical insights into:

- Engine characteristics influencing fuel type selection
- Customer preference patterns
- Performance vs affordability segmentation
- Pricing alignment with fuel types

These insights are directly relevant for strategic product planning as Geely Auto prepares to enter the US market.

---

7) Conclusion of Dataset Overview

The dataset is structured, clean, and business-relevant. It provides sufficient technical and economic variables to build a predictive model for fuel type selection. The presence of meaningful engine and efficiency-related variables makes it highly suitable for Logistic Regression analysis and strategic decision-making.